

Precipitation data is available from the Bureau of Meteorology website as daily, monthly or annual data. Gridded data for the Australian continent starting from 2005, in netCDF format, can be downloaded. Download the monthly data netCDF, file for all available years, from the website:

<https://awo.bom.gov.au/>

1. (4 marks) plot the Actual rainfall data for the following months (as four subplots in one figure):
 - January 2005
 - August 2009
 - February 2010
 - December 2019
2. (4 marks) Derive the integrated rainfall that fell each month within the Murray-Darling Basin. The file MDB.latlon contains a polygon defining the spatial extent of the Murray-Darling Basin and is available from [here](#).
3. (1 mark) Plot a time series of the monthly rainfall, integrated over the Murray-Darling Basin.
4. (1 mark) Calculate the rate of change in monthly rainfall over the time span of the data.
5. (4 marks) Compare the drought periods of 2005-2009 and 2016-2019 in terms of precipitation. What was similar and what was different about these two droughts?
6. (4 marks) Compare the la Nina periods of 2010-2012 with those of 2020-2023. What was similar and different about these periods? (For more information on El Niño and La Niña click [here](#)).

Note:

the “time” variable in the BoM precipitation netCDF files refers to the number of days since 01/01/1900